



```

import A5O
import Closure
import Deployment
import DeploymentTests
import ControlPlane
set_option autoImplicit false

```

```

/-! # A5O — the apertures.app IS a Control Plane (checked import).

```

Imports, unmodified:

```

A5O.lean v4      0e5ebe3f...5b452
Closure.lean     af6b3233...a933
Deployment.lean v5 e07c69ae...59da (live-gate deployment model of record)
DeploymentTests.lean 1386ffb1...24d8 (v5 synthetic fixtures)
ControlPlane.lean 8d7c36cf...c463d

```

v5 builds the live gate as a `GatedSystem` from a trace (`theApertures`) and proves it achieves ADA (`gate\_achievesADA`). Here those two v5 objects are assembled into a `ControlPlane`, and every ControlPlane theorem is applied to the installation — over the logged trace and over ACTUAL executions. The four v5 obligations stay explicit hypotheses:

```

  CheckersSound, CoversActualExecutions — the signed operator attestations;
  DecisionsFollowChecker, NoSideChannel — decided over the signed trace.

```

No obligation is discharged or weakened in this file. -/

```

namespace Apertures

```

```

/-! ## The installation as a control plane -/

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```

/-- Over the logged trace: the v5 gate, as a control plane. -/

```

```

def aperturesPlane (s : EvidenceSpec) (v : Checkers) (d : Deployment)
  (hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
  (hc : NoSideChannel d) :
  ControlPlane s.C s.I s.M s.S s.R s.Appointed (Executes d) inAoC where
gate := theApertures d hc
ada  := gate_achievesADA s v d hs hd

```

```

/-- Over what ACTUALLY executes: the gate's interception extends from the
trace to real executions through CoversActualExecutions. -/

```

```

def aperturesPlaneActual (s : EvidenceSpec) (v : Checkers) (d : Deployment)
  (actual : Agent → Action → Time → Prop)
  (hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
  (hc : NoSideChannel d) (ha : CoversActualExecutions d actual) :
  ControlPlane s.C s.I s.M s.S s.R s.Appointed actual inAoC where

```

```

gate :=
  { sys := gate d
    intercepts := fun a x t dom hDom hAct =>
      gate_intercepts d hc a x t dom hDom (ha a x t hAct) }
ada := gate_achievesADA s v d hs hd

```

```

/-- The plane over actual executions uses the same decision function as
the v5 gate — nothing is added between the trace and the plane. -/
theorem aperturesPlaneActual_is_v5_gate (s : EvidenceSpec) (v : Checkers)
  (d : Deployment) (actual : Agent → Action → Time → Prop)
  (hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
  (hc : NoSideChannel d) (ha : CoversActualExecutions d actual) :
  (aperturesPlaneActual s v d actual hs hd hc ha).gate.sys = gate d := rfl

```

```

/-! ## The seven ControlPlane properties, at the installation -/

```

```

/-- (1) Every actual consequential execution was decided by the gate — an
issued ALLOW receipt exists — and its principal satisfies O. -/
theorem apertures_decides_every_actual_execution
  (s : EvidenceSpec) (v : Checkers) (d : Deployment)
  (actual : Agent → Action → Time → Prop)
  (hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
  (hc : NoSideChannel d) (ha : CoversActualExecutions d actual)
  (a : Agent) (x : Action) (t : Time) (hAct : actual a x t) :
  ∃ p : Principal, (gate d).enforces p a x t ∧
  O s.C s.I s.M s.S s.R s.Appointed p a x t :=
plane_decides_every_execution s.C s.I s.M s.S s.R s.Appointed actual inAoC
  (aperturesPlaneActual s v d actual hs hd hc ha) a x t () trivial hAct

```

```

/-- (1') Accountability over actual executions, now obtained through the
control-plane theorem (agrees with v5's `apertures_accountable`). -/
theorem apertures_plane_accountable
  (s : EvidenceSpec) (v : Checkers) (d : Deployment)
  (actual : Agent → Action → Time → Prop)
  (hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
  (hc : NoSideChannel d) (ha : CoversActualExecutions d actual) :
  Accountable s.C s.I s.M s.S s.R s.Appointed actual inAoC :=
plane_accountable s.C s.I s.M s.S s.R s.Appointed actual inAoC
  (aperturesPlaneActual s v d actual hs hd hc ha)

```

```

/-- (2) Single governor at the installation, under CFunctional of the spec. -/
theorem apertures_single_governor
  (s : EvidenceSpec) (v : Checkers) (d : Deployment)
  (actual : Agent → Action → Time → Prop)

```

```

(hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
(hc : NoSideChannel d) (ha : CoversActualExecutions d actual)
(hC : CFunctional s.C)
(a : Agent) (x : Action) (t : Time) (hAct : actual a x t) :
  ∃ p : Principal, (gate d).enforces p a x t ∧
    ∀ p' : Principal, (gate d).enforces p' a x t → p' = p :=
plane_single_governor s.C s.I s.M s.S s.R s.Appointed actual inAoC hC
(aperturesPlaneActual s v d actual hs hd hc ha) a x t () trivial hAct

```

/-- (3) No rival: ANY control plane over the same evidence spec that decides the same act names the same principal as the apertures.app gate. -/

theorem apertures_no_rival_governor

```

(s : EvidenceSpec) (v : Checkers) (d : Deployment)
(actual : Agent → Action → Time → Prop)
(hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
(hc : NoSideChannel d) (ha : CoversActualExecutions d actual)
(hC : CFunctional s.C)
(rival : ControlPlane s.C s.I s.M s.S s.R s.Appointed actual inAoC)
(p q : Principal) (a : Agent) (x : Action) (t : Time)
(hA : (gate d).enforces p a x t) (hR : rival.gate.sys.enforces q a x t) :
  p = q :=
planes_agree_on_governor s.C s.I s.M s.S s.R s.Appointed actual inAoC hC
(aperturesPlaneActual s v d actual hs hd hc ha) rival p q a x t () trivial hA hR

```

/-- (4) One form: every ALLOW the installation issues lies inside the canonical O monitor. -/

theorem apertures_within_canonical

```

(s : EvidenceSpec) (v : Checkers) (d : Deployment)
(actual : Agent → Action → Time → Prop)
(hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
(hc : NoSideChannel d) (ha : CoversActualExecutions d actual)
(p : Principal) (a : Agent) (x : Action) (t : Time)
(hEnf : (gate d).enforces p a x t) :
  (CanonicalMonitor s.C s.I s.M s.S s.R s.Appointed).enforces p a x t :=
plane_within_canonical s.C s.I s.M s.S s.R s.Appointed actual inAoC
(aperturesPlaneActual s v d actual hs hd hc ha) p a x t () trivial hEnf

```

/-- (5) Revocation halts the act at the installation: after the named principal revokes, the act does not actually execute. -/

theorem apertures_revocation_halts

```

(s : EvidenceSpec) (v : Checkers) (d : Deployment)
(actual : Agent → Action → Time → Prop)
(hs : CheckersSound s v) (hd : DecisionsFollowChecker v d)
(hc : NoSideChannel d) (ha : CoversActualExecutions d actual)

```

```

(Revoke : Principal → Agent → Action → Time → Prop)
(hP2 : ∀ (p : Principal) (a : Agent) (x : Action) (t t' : Time),
  t < t' → s.Appointed p a x t → Revoke p a x t' → ¬ s.R p a x t')
(hC : CFunctional s.C)
(p : Principal) (a : Agent) (x : Action) (t t' : Time)
(hlt : t < t') (hApp : s.Appointed p a x t) (hRev : Revoke p a x t')
(hBind : s.C p a x t') :
  ¬ actual a x t' :=
revocation_halts_execution s.C s.I s.M s.S s.R s.Appointed actual inAoC
  Revoke hP2 hC (aperturesPlaneActual s v d actual hs hd hc ha)
  p a x t t' () trivial hlt hApp hRev hBind

/-! ## Non-vacuity at the v5 fixture: a populated installation-shaped plane -/

open Tests in
theorem spec_CFunctional : CFunctional spec.C :=
  fun _ _ _ _ h1 h2 => h1.1.trans h2.1.symm

open Tests in
/-- (7) On the v5 synthetic trace (one ALLOW executed, one revocation DENY),
all four obligations are discharged by v5's own fixture theorems; the plane
exists, the execution occurs, and it has exactly one governor. -/
theorem demo_plane_single_governor :
  Executes demo "demo-agent" "brief" 10 ∧
  ∃ p : Principal, (gate demo).enforces p "demo-agent" "brief" 10 ∧
  ∀ p' : Principal, (gate demo).enforces p' "demo-agent" "brief" 10 → p' = p :=
⟨allow_executed,
plane_single_governor spec.C spec.I spec.M spec.S spec.R spec.Appointed
  (Executes demo) inAoC spec_CFunctional
  (aperturesPlane spec checks demo checks_sound demo_decisions demo_coverage)
  "demo-agent" "brief" 10 () trivial allow_executed⟩

end Apertures

#print axioms Apertures.aperturesPlaneActual_is_v5_gate
#print axioms Apertures.apertures_decides_every_actual_execution
#print axioms Apertures.apertures_plane_accountable
#print axioms Apertures.apertures_single_governor
#print axioms Apertures.apertures_no_rival_governor
#print axioms Apertures.apertures_within_canonical
#print axioms Apertures.apertures_revocation_halts
#print axioms Apertures.spec_CFunctional
#print axioms Apertures.demo_plane_single_governor

```